COMPUTER *vision*

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ASSIGMENT-I

TRYING DIFFERENT ACTIVATION FUNCTIONS

MY PROGRESS

This project focuses on building an Artificial Neural Network (ANN) model to predict employee attrition using the IBM HR Analytics Employee Attrition Dataset.
First, I performed Exploratory Data Analysis (EDA) to understand the dataset, including checking data types, summary statistics, and the distribution of the target variable (Attrition). This helped in identifying patterns and ensuring data quality.

Next, I carried out data preprocessing, which included removing irrelevant columns, encoding categorical variables into numerical form, and splitting the dataset into training and testing sets.

CONTENT

After preparing the data, I implemented an ANN using TensorFlow. The model consists of an input layer, hidden layers, and an output layer designed for binary classification.

A key part of this project was experimenting with different activation functions (ActFn), as mentioned below:

- Sigmoid
- Tanh
- ReLU

TheseActFn were applied in the hidden layers to analyze their impact on model performance.

Additionally, I performed hyperparameter tuning, where I experimented with:

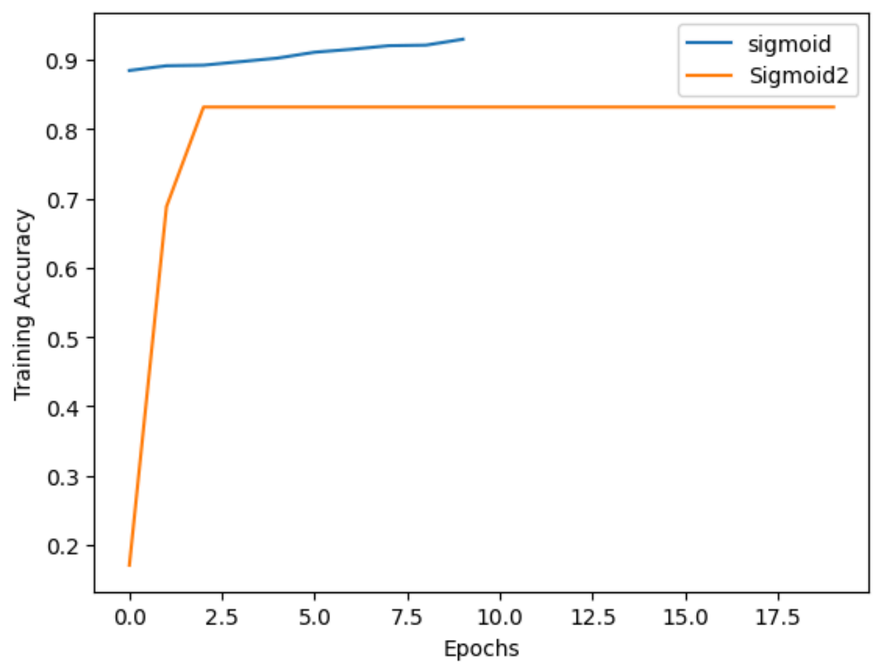
- Different learning rates (LR)
- Various batch sizes
- Multiple epoch values

This helped in optimizing the model's performance and achieving better accuracy (Acc).

Finally, I evaluated the models and compared their performance. It was observed that ReLU activation function provided the best results, due to faster training and better handling of the vanishing gradient problem.

SIGMOID

Sigmoid is an activation function used in neural networks.
It takes any number and converts it into a value between 0 and 1.
$\sigma(x)=1/(1+\exp(-x))$



SIGMOID 1

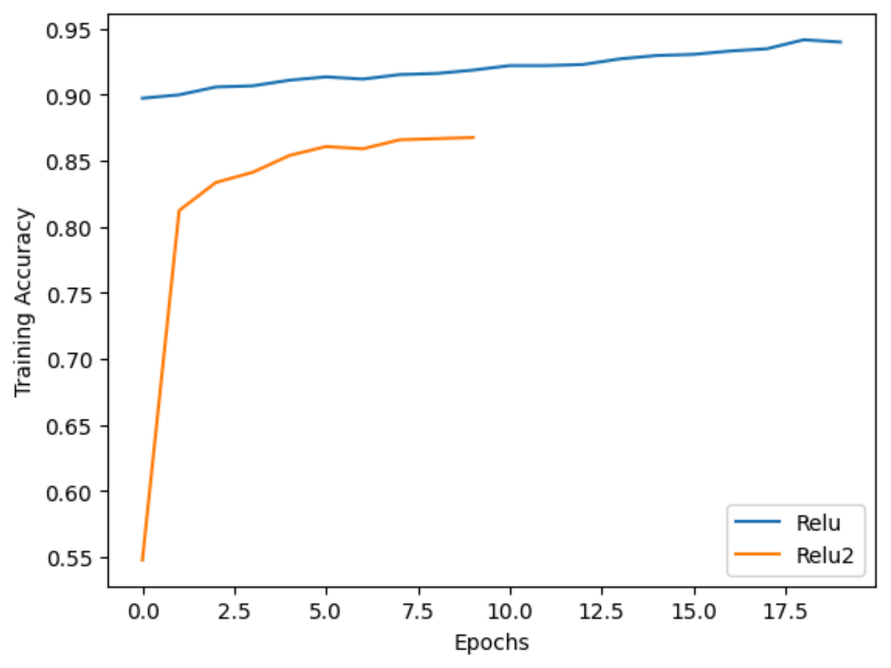
EPOCHS=20,
BATCH_SIZE=16
LEARNING RATE -PREFIXED

ACC: 0.884353

SIGMOID 2

EPOCHS=20,
BATCH_SIZE=64
LEARNING_RATE=0.001

ACCURACY: 0.867346



RELU

RELU 1

EPOCHS=20,
BATCH_SIZE=16
LEARNING RATE -PREFIXED

ACCURACY: 0.8639

RELU 2

EPOCHS=10,
BATCH_SIZE=34
LEARNING_RATE=0.001

ACCURACY: 0.8741

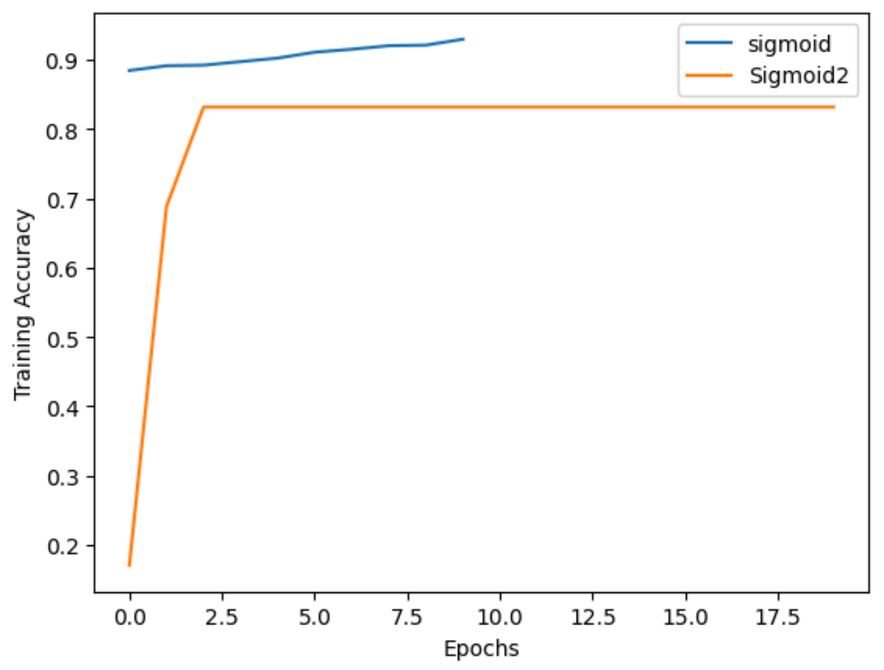
It is an activation function used in neural networks that outputs:

- the same value if it's positive
- 0 if it's negative

f(x)=max(0,x)

TANH

Tanh = Hyperbolic Tangent function
It is an activation function that converts input values into a range between -1 and +1



TANH 1

EPOCHS=20,
BATCH_SIZE=36
LEARNING RATE -PREFIXED

ACCURACY: 0.8877

TANH 2

EPOCHS=5,
BATCH_SIZE=64
LEARNING_RATE=0.01

ACCURACY: 0.9113

RESULT AND CONCLUSION

WE CAN NOTICE THAT ALL THE RESULTS ARE ALMOST SIMILAR THERE ARE NO DRASTIC CHNAGE BUT WE SEE THERE IS CHANGES WHEN WE CHANGE MULTIPLE FACTORS LIKE BATCH SIZE,EPOCHS,LEARNING RATE SO MY CONCLUSION IS THAT ITS GOOD TO TRY DIFFERENT FACTORS AS WE CAN SEE CHANGING FACTORS GIVEN BETTER ACCURACY THEN PREVIOUS MODEL SOME HAVE TAKEN LESS TIME

Serieal No.	Activation Function	Parameter	Accuracy
1	SIGMOID	epochs=20, batch_size=16 Learning rate -prefixed	0.884353
		epochs=20, batch_size=64 learning_rate=0.001	0.867346
2	RELU	epochs=20, batch_size=16 Learning rate -prefixed	0.8912
		epochs=10, batch_size=34 learning_rate=0.001	0.8912
3	TANH	epochs=20, batch_size=36 Learning rate -prefixed	0.8877
		epochs=5, batch_size=64 learning_rate=0.01	0.9113

ASSIGMENT-2

WORKING WITH CIFAR-10

INTRODUCTION

- Convolutional Neural Networks (CNNs) are a class of deep learning models used for image processing
- They automatically extract important features from images
- Widely used in:
  - Image classification
  - Object detection
  - Face recognition
- In this project, we use the CIFAR-10 dataset for classification

Dataset Info:

- 60,000 images (32×32 pixels)
- 10 classes: airplane, car, bird, cat, deer, dog, frog, horse, ship, truck

MODEL SUMMARY

Activation functions help the model learn complex patterns.

1. ReLU (Rectified Linear Unit)

Formula: f(x) = max(0, x)

Fast and efficient

Most commonly used

2. Sigmoid

Output between 0 and 1

Causes vanishing gradient problem

3. Tanh

Output between -1 and 1

Better than sigmoid but slower

- Input Layer: Image (32×32×3)
- Convolution Layers:
  - Extract features using filters/kernels
- Activation Function:
  - Adds non-linearity
- Pooling Layer:
  - Reduces image size and computation
- Fully Connected Layer:
  - Final classification
- Output Layer:
  - 10 classes using Softmax

Key Layers Used:

- Conv2D
- MaxPooling2D
- Dense
- Flatten

EXPERIMENTS

Dataset: CIFAR-10

- Optimizer: Adam
- Loss Function: Sparse Categorical Crossentropy
- Epochs: 10
- Experiments Performed:

Changed activation functions:

- ReLU
- Sigmoid
- Tanh
- Compared model performance

SIGMOID

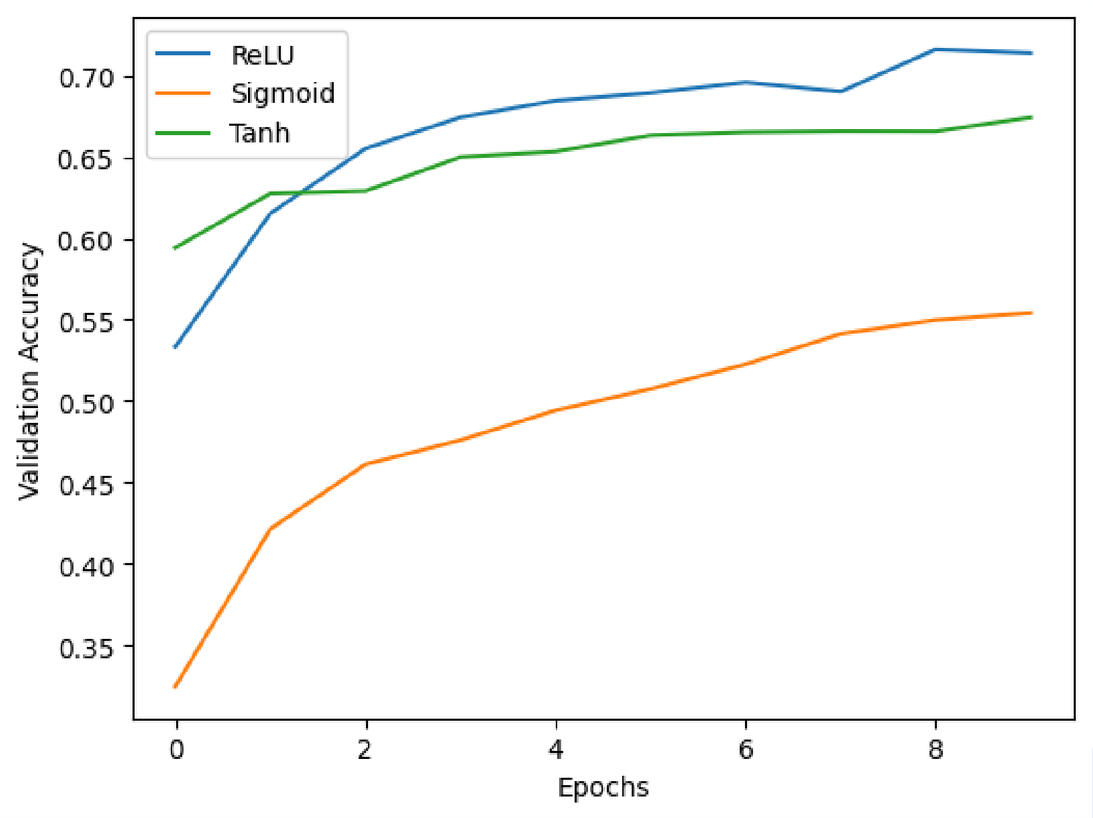
RELU

TANH

SIGMOID ACCURACY: 0.55419

RELU ACCURACY: 0.7135

TANH ACCURACY: 0.6740



CONCLUSION

CNN effectively classifies images using feature extraction

Activation functions play a crucial role in performance

- ReLU is the best choice for deep CNN models
- Sigmoid and Tanh are less efficient for deep networks

Future Work:

- Use Dropout to reduce overfitting
- Try deeper networks
- Use advanced models like ResNet

THANK YOU

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